

Software Requirements Specification (SRS)

1. DOCUMENT INFORMATION

Field	Details
Project Name	CampusAwaz
Document Title	Software Requirements Specification
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2. INTRODUCTION

2.1 Purpose

CampusAwaz is an AI-powered, privacy-first student complaint and support platform designed to connect verified university students and eligible alumni with university administrations. This SRS defines the functional, non-functional, UI/UX, integration, architecture, security, notification, analytics, and acceptance requirements for the CampusAwaz MVP.

2.2 Scope

The system will allow students from participating universities to verify their university affiliation, submit university-related complaints, choose their preferred level of identity protection, upload evidence, receive AI assistance, track complaints, and receive resolution information.

University authorities will be able to manage, assign, review, escalate, and resolve complaints through role-based dashboards. Sensitive harassment and safety complaints will use a protected workflow with restricted access.

The MVP will focus on a responsive web application and a complete complaint-to-resolution workflow.

4. PRODUCT SCOPE

4.1 In Scope

- University and student registration
- Current student and graduate status
- University affiliation verification
- University email/OTP verification
- Student-card verification with AI/OCR assistance
- Verified but Protected reporting
- Identified, Confidential, and Anonymous complaint options
- Universal complaint system
- Academic and marking complaints
- Attendance and examination complaints
- Harassment, bullying, ragging and stalking complaints
- Wi-Fi and infrastructure complaints
- Hostel and transport complaints
- Fee and scholarship complaints
- Administrative, degree and transcript complaints
- Evidence upload
- AI Complaint Assistant
- AI complaint classification
- AI priority recommendation
- AI department recommendation
- AI structured complaint generation
- Smart complaint routing
- Dedicated harassment and safety workflow
- Female Focal Person access

- Immediate-danger indication
- Complaint tracking ID
- Complaint status timeline
- Complaint escalation
- Proof of Action
- Resolution evidence
- Student feedback
- Student support resources
- Mental-health resources
- Anonymous counseling requests
- Student-rights and policy information
- FAQs
- Emergency contacts
- University administration dashboard
- Role-based access
- Complaint analytics
- Notifications for complaint status changes
- Audit logging
- Responsive web interface

4.2 Out of Scope

- Official live university database integration during the MVP
- Official HEC integration during the MVP
- SMS services during the MVP
- Real-time campus security tracking
- Large-scale production deployment across universities

4.3 Future Enhancements

- Official university student-information-system integrations
 - HEC integration where official APIs are available
 - Automatic escalation to external bodies
 - SMS services
 - Large-scale multi-university deployment
 - Dedicated mobile application
 - Additional student-community features
 - Advanced analytics
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7. FUNCTIONAL REQUIREMENTS

7.1 University & Student Authentication Module

Purpose

To register users and verify their affiliation with a participating university.

Actors

- Student
- Graduate/Alumni
- System

Preconditions

- User must access the registration process.
- The selected university must be available in the system.

Main Flow

1. User creates an account.
2. User selects a university.
3. User selects Current Student or Graduate.
4. User completes an available verification method.
5. System records the verification status.

6. Verified user can access complaint functionality.

Alternative Flows

- User may verify through university email/OTP.
- User may use student-card verification where available.

Error/Exception Scenarios

- Verification information does not match.
- Invalid or incomplete verification information is provided.
- Verification cannot be completed.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-AUTH-001	The system shall allow users to select their participating university.	High	University can be selected during registration.
FR-AUTH-002	The system shall allow users to identify themselves as a current student or graduate.	High	User can select one applicable status.
FR-AUTH-003	The system shall support university email/OTP verification.	High	User can complete verification using the supported university email method.
FR-AUTH-004	The system shall support student-card verification where enabled.	High	Student can submit a student card for verification.
FR-AUTH-005	The system shall record the user's university verification status.	High	Verification status is available after the verification process.

7.2 Student Complaint Module

Purpose

To provide one centralized system for reporting university-related problems.

Actors

- Verified Student
- Graduate/Alumni

Preconditions

- User must be verified.

Main Flow

1. Student opens the complaint system.
2. Student describes the problem.
3. Student selects or confirms the complaint category.
4. Student selects identity protection level.
5. Student attaches evidence if available.
6. Student reviews the complaint.
7. Student submits it.
8. System generates a unique tracking ID.

Alternative Flows

- Student may submit without evidence.
- Student may submit as Identified, Confidential, or Anonymous.

Error/Exception Scenarios

- Required complaint information is missing.
- Evidence upload fails.
- Complaint cannot be submitted.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-CMP-001	The system shall allow verified users to submit university-related complaints.	High	A verified user can successfully submit a complaint.
FR-CMP-002	The system shall support academic, facilities, hostel, financial, administrative, safety, harassment, and other university-related complaint categories.	High	Supported categories are available when creating a complaint.
FR-CMP-003	The system shall allow students to attach supporting evidence.	High	Supported evidence can be attached to a complaint.
FR-CMP-004	The system shall provide Identified, Confidential, and Anonymous reporting options.	High	Student can select one privacy option before submission.
FR-CMP-005	The system shall generate a unique tracking ID for each submitted complaint.	High	A unique ID is generated after successful submission.

7.3 AI Complaint Assistant

Purpose

To help students describe and structure complaints using natural language.

Actors

- Verified Student
- AI Service

Preconditions

- Student must access the complaint assistant.

Main Flow

1. Student describes the problem naturally.
2. AI analyzes the description.
3. AI recommends category, issue, priority, and department.
4. AI can generate a structured complaint.
5. Student reviews the result.
6. Student submits or modifies the complaint.

Alternative Flows

- Student may modify the AI-generated complaint.
- Administrator may review and modify AI recommendations.

Error/Exception Scenarios

- AI cannot determine a category.
- AI service is unavailable.
- AI-generated information requires correction.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-AI-001	The system shall allow students to describe complaints in natural language.	High	Natural-language complaint text can be entered.
FR-AI-002	AI shall recommend a complaint category based on the description.	High	A category recommendation is returned when AI processing succeeds.
FR-AI-003	AI shall recommend complaint priority where sufficient information is available.	High	A priority recommendation is displayed.

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-AI-004	AI shall recommend a responsible department where sufficient information is available.	High	A department recommendation is displayed.
FR-AI-005	AI shall be able to convert an informal description into a structured complaint.	High	A structured complaint can be generated for student review.
FR-AI-006	The student shall be able to review AI-generated complaint information before submission.	High	Student can edit or approve the generated information.

7.4 Harassment & Safety Module

Purpose

To provide a protected reporting flow for harassment and safety-related complaints.

Actors

- Student
- Female Focal Person
- Authorized University Staff
- System

Preconditions

- Student must have access to the complaint system.

Main Flow

1. Student selects a harassment or safety complaint.
2. System opens the protected workflow.
3. Student describes the incident.
4. Student selects Anonymous or Confidential reporting if required.

5. Student indicates whether there is immediate danger.
6. Student uploads evidence if available.
7. Complaint is routed to authorized personnel.
8. Authorized personnel review the case.

Alternative Flows

- Student may submit an identified report.
- Student may submit without evidence.
- Case may be routed to a Female Focal Person.

Error/Exception Scenarios

- Unauthorized user attempts to access sensitive information.
- Evidence upload fails.
- Sensitive complaint cannot be routed.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-HAR-001	The system shall provide a dedicated harassment and safety reporting workflow.	High	Harassment/safety complaints use the protected workflow.
FR-HAR-002	The system shall support anonymous and confidential harassment reporting.	High	Student can select protected reporting.
FR-HAR-003	The system shall allow students to indicate whether they are in immediate danger.	High	Immediate-danger information can be submitted with the case.
FR-HAR-004	The system shall allow harassment evidence to be submitted.	High	Supporting evidence can be attached.

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-HAR-005	The system shall route sensitive cases to authorized personnel.	High	Unauthorized administrators cannot access restricted cases.
FR-HAR-006	The system shall support a Female Focal Person role for sensitive cases.	High	Authorized Female Focal Person can access assigned cases.

7.5 Smart Routing & Priority Module

Purpose

To assist university authorities in directing complaints to the appropriate department or authority.

Actors

- AI Service
- University Administrator
- Authorized Staff

Preconditions

- Complaint must be submitted.

Main Flow

1. AI analyzes complaint information.
2. AI recommends category, priority, and department.
3. Administrator reviews the recommendations.
4. Administrator modifies them if required.
5. Complaint is assigned.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-ROUTE-001	The system shall provide AI-based routing recommendations.	High	Recommended department is displayed.
FR-ROUTE-002	The system shall provide AI-based priority recommendations.	High	Recommended priority is displayed.
FR-ROUTE-003	Authorized administrators shall be able to review AI recommendations.	High	Administrator can view recommendations.
FR-ROUTE-004	Authorized administrators shall be able to modify AI recommendations before assignment.	High	Administrator can change the recommendation before assigning the complaint.

7.6 Complaint Tracking Module

Purpose

To allow students to monitor the progress of submitted complaints.

Actors

- Student
- University Administrator

Main Flow

1. Student opens a complaint using its tracking ID.
2. System displays the complaint status.
3. System displays the status timeline.
4. Student receives updates when relevant information changes.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-TRACK-001	The system shall provide a unique tracking ID for every complaint.	High	Each complaint has an identifiable tracking ID.
FR-TRACK-002	The system shall display complaint status.	High	Current complaint status is visible.
FR-TRACK-003	The system shall display the complaint timeline.	High	Status history is displayed chronologically.
FR-TRACK-004	The system shall support Submitted, Assigned, In Review, Action Taken, and Resolved statuses.	High	Supported statuses can be applied to complaints.
FR-TRACK-005	The system shall support Escalated and Reopened states.	Medium	Authorized users can use applicable states.

7.7 Escalation Module

Purpose

To ensure unattended complaints can move to higher authorities.

Actors

- System
- University Administrator
- HOD
- Proctor
- Higher Administration

Preconditions

- Complaint must remain unattended beyond the defined response period.

Main Flow

1. System monitors the complaint response period.

2. Response period is exceeded.
3. Complaint enters escalation workflow.
4. Case moves to the next responsible authority.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-ESC-001	The system shall support configurable complaint response periods.	High	Response period can be defined for escalation.
FR-ESC-002	The system shall identify complaints exceeding their defined response period.	High	Eligible complaints are identified for escalation.
FR-ESC-003	The system shall support escalation to higher university authorities.	High	An eligible complaint can be escalated.
FR-ESC-004	The system shall record escalation activity.	High	Escalation information is stored with the complaint.

7.8 Proof of Action & Resolution Module

Purpose

To make complaint resolution transparent by requiring administrators to document actions taken.

Actors

- University Administrator
- Authorized Staff
- Student

Main Flow

1. Authorized administrator investigates the complaint.
2. Administrator records the action taken.

3. Administrator provides resolution details.
4. Administrator uploads supporting proof where available.
5. Complaint is marked Resolved.
6. Student views the resolution and proof.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-RES-001	The system shall allow authorized administrators to record actions taken.	High	Action details can be entered.
FR-RES-002	The system shall allow resolution evidence to be uploaded.	High	Supporting proof can be attached.
FR-RES-003	The system shall provide resolution information to the student.	High	Student can view the resolution record.
FR-RES-004	The system shall maintain a Proof of Action record for resolved complaints.	High	Resolved complaint contains the recorded action information.
FR-RES-005	Students shall be able to provide feedback after resolution.	Medium	Feedback can be submitted after resolution.

7.9 University Administration Module

Purpose

To provide university authorities with centralized complaint management.

Actors

- University Administrator
- HOD
- Proctor

- Female Focal Person
- Hostel Warden
- Counselor
- Other Authorized Staff

Main Flow

1. Authorized staff accesses the dashboard.
2. Staff views relevant complaints.
3. Staff reviews AI recommendations.
4. Staff assigns and manages cases.
5. Staff updates status.
6. Staff escalates or resolves cases where applicable.

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-ADMIN-001	The system shall provide an administration dashboard.	High	Authorized staff can access the dashboard.
FR-ADMIN-002	The system shall allow authorized staff to view relevant complaints.	High	Staff can view complaints according to their permissions.
FR-ADMIN-003	The system shall allow authorized staff to assign complaints.	High	A complaint can be assigned to an appropriate authority.
FR-ADMIN-004	The system shall allow authorized staff to update complaint status.	High	Authorized staff can update applicable statuses.
FR-ADMIN-005	The system shall restrict sensitive complaints according to assigned roles.	High	Unauthorized staff cannot access restricted cases.

7.10 Student Support Module

Purpose

To provide students with support and relevant information in addition to complaint reporting.

Actors

- Student
- System

Functional Requirements

Requirement ID	Requirement	Priority	Acceptance Criteria
FR-SUP-001	The system shall provide mental-health support resources.	Medium	Available resources can be accessed by students.
FR-SUP-002	The system shall support anonymous counseling requests.	Medium	A student can submit a counseling request anonymously where supported.
FR-SUP-003	The system shall provide student-rights information.	Medium	Student-rights information is accessible.
FR-SUP-004	The system shall provide university policies and FAQs.	Medium	Available information can be accessed by students.
FR-SUP-005	The system shall provide emergency contacts.	Medium	Emergency contact information is accessible.

11. NON-FUNCTIONAL REQUIREMENTS

11.1 Performance

No specific numerical performance targets were provided.

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-PERF-001	The system should provide responsive interaction for normal student and administrative operations.	Medium	Core operations function without unnecessary delays under normal prototype usage.

11.2 Security

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-SEC-001	The system shall restrict functionality according to user roles.	High	Users cannot access functions outside their permissions.
NFR-SEC-002	Sensitive complaint information shall only be accessible to authorized personnel.	High	Unauthorized users cannot access restricted information.
NFR-SEC-003	The system shall validate user input and uploaded evidence.	High	Invalid input or unsupported uploads are handled appropriately.
NFR-SEC-004	The system shall maintain audit information for relevant sensitive actions.	High	Relevant actions can be traced through audit records.

11.3 Reliability

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-REL-001	Complaint records and status information shall remain available after successful submission.	High	Submitted complaints can be retrieved through the system.

11.4 Availability

Specific uptime requirements are not specified.

11.5 Scalability

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-SCAL-001	The system architecture shall support future expansion to additional universities and integrations.	Medium	The design does not restrict the system to a single university.

11.6 Usability

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-USE-001	The complaint process shall allow students to describe problems in natural language.	High	Students can begin a complaint without understanding complex procedures.
NFR-USE-002	Complaint progress shall be presented through a simple status timeline.	High	Students can understand the current stage of a complaint.

11.7 Accessibility

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-ACC-001	The system shall consider accessibility for students with disabilities.	Medium	Core interface elements are designed with accessibility considerations.

Specific accessibility standards are not specified.

11.8 Maintainability

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-MAIN-001	The application shall use a modular structure to support future feature development.	Medium	Core modules can be maintained and extended independently.

11.9 Compatibility

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-COMP-001	The web application shall support desktop, tablet, and mobile browser layouts.	High	Core screens remain usable across supported screen sizes.

11.10 Privacy

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-PRIV-001	Verified students shall be able to submit complaints without automatically exposing their identity to unauthorized complaint handlers.	High	Anonymous/confidential complaints do not expose identity to unauthorized users.
NFR-PRIV-002	Sensitive harassment information shall have restricted access.	High	Only authorized roles can access protected information.
NFR-PRIV-003	The system shall follow a privacy-first approach for student complaint information.	High	Personal information is only exposed where required by the selected reporting mode and user permissions.
NFR-PRIV-004	Data is intended to be hosted in Pakistan as specified for the project.	Medium	Deployment follows the project's stated data-hosting requirement

Requirement ID	Requirement	Priority	Acceptance Criteria
			where supported by the selected infrastructure.

11.11 Backup and Recovery

Backup schedules and recovery targets are not specified.

12. UI/UX REQUIREMENTS

Main Screens/Pages

The system will provide UI for:

- Landing page
- Registration
- University selection
- Verification
- Student dashboard
- AI Complaint Assistant
- Complaint submission
- Privacy selection
- Evidence upload
- Complaint tracking
- Harassment and safety reporting
- Student support
- Admin dashboard
- Complaint management
- Complaint details
- Resolution and Proof of Action

- Analytics
- Female Focal Person/sensitive-case interface

Navigation

The interface shall provide clear navigation between available student or administrative functions according to the user's role.

Forms

Complaint forms shall support:

- Natural-language description
- Complaint category
- Privacy selection
- Evidence upload
- Harassment/safety information where applicable
- Review before submission

Validation and States

The system shall provide appropriate:

- Input validation
- Error states
- Loading states
- Empty states
- Success states

Exact UI messages are not specified.

Responsive Behavior

The web application shall be responsive for desktop, tablet, and mobile screens.

Accessibility

The interface shall consider accessibility for students with disabilities.

13. API / INTEGRATION REQUIREMENTS

The following integrations are identified in the project information:

Integration	Purpose	Current MVP Status
University email/OTP	Student affiliation verification	Applicable
Student-card OCR/Vision	Extract information from student cards	Applicable
University student database	Validate student affiliation	Future
HEC integration	External escalation	Future
SMS services	Notifications	Future

Specific API endpoints, request fields, response fields, and external API specifications are not specified.

The AI service will support complaint classification, priority recommendation, department recommendation, structured complaint generation, and student assistance.

14. SYSTEM ARCHITECTURE

The MVP will use the following high-level architecture:

Student/Admin Interface

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Next.js Application / API Layer

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Supabase PostgreSQL + Storage

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AI/OCR Services

Frontend

- Next.js
- React
- TypeScript
- Tailwind CSS

Backend

- Next.js API/server-side functions

Database and Storage

- Supabase PostgreSQL
- Supabase Storage

AI Services

AI/LLM services will support complaint understanding, classification, prioritization, routing, structured complaint generation, and student assistance.

OCR/Vision capabilities will assist with student-card verification.

Authentication

Authentication and university verification will be supported through the application's authentication and verification flow.

Data Flow

Student registration

- University verification
- Complaint description
- AI analysis
- Privacy selection
- Evidence
- Complaint submission
- Smart routing
- Administrator action
- Tracking
- Escalation where required
- Proof of Action
- Resolution

16. NOTIFICATIONS

Notification/Event	Trigger	Recipient	Channel	Message/Purpose
Complaint status update	Complaint status changes	Student	Not specified	Inform student about complaint progress

Notification/Event	Trigger	Recipient	Channel	Message/Purpose
Case information update	Relevant case information changes	Student	Not specified	Keep student informed
Escalation	Complaint exceeds response period	Responsible authority	Not specified	Inform authority of escalated case
Resolution	Complaint is resolved	Student	Not specified	Inform student of resolution and Proof of Action

Specific notification channels are not specified for the MVP.

17. REPORTING / ANALYTICS

The university administration dashboard shall provide aggregated analytics for identifying complaint patterns.

Supported analytics include:

- Complaints by category
- Complaints by department
- Average resolution time
- Pending complaints
- Escalated complaints
- Resolution rate
- Recurring facility problems

Analytics shall avoid exposing sensitive personal information unnecessarily.

Export functionality is not specified.

21. SECURITY REQUIREMENTS

Requirement ID	Requirement	Priority	Acceptance Criteria
NFR-SEC-005	The system shall authenticate users before providing protected functionality.	High	Unauthenticated users cannot access protected student/admin functions.
NFR-SEC-006	The system shall implement role-based authorization.	High	Users can access only functions permitted for their role.
NFR-SEC-007	The system shall protect anonymous complaint identity from unauthorized users.	High	Complaint handlers without authorization cannot view anonymous identity information.
NFR-SEC-008	The system shall restrict sensitive harassment cases to authorized roles.	High	Unauthorized administrators cannot access protected harassment information.
NFR-SEC-009	The system shall validate complaint inputs and uploaded evidence.	High	Invalid or unsupported information is rejected or handled appropriately.
NFR-SEC-010	The system shall maintain audit records for relevant sensitive actions.	High	Authorized access and relevant case actions can be traced.

Specific encryption algorithms, password policies, session durations, and security standards are not specified.

22. ACCEPTANCE CRITERIA

CampusAwaz will be considered acceptable for the MVP when:

1. A student can register and select a participating university.
2. A student can verify their university affiliation through the supported verification flow.
3. A verified student can submit a university-related complaint.
4. The student can choose Identified, Confidential, or Anonymous reporting.

5. The student can attach supporting evidence.
6. The AI Complaint Assistant can analyze a complaint and provide applicable recommendations.
7. AI-generated complaint information can be reviewed before submission.
8. Harassment and safety complaints use a protected workflow.
9. Sensitive cases are restricted to authorized personnel.
10. Every submitted complaint receives a tracking ID.
11. Students can view complaint progress through the defined status timeline.
12. Administrators can view and manage complaints according to their roles.
13. Administrators can assign and update complaints.
14. Complaints can enter the escalation workflow when the defined response period is exceeded.
15. Administrators can provide resolution information and Proof of Action.
16. Students can view the resolution information.
17. Relevant complaint status changes can generate notifications.
18. The system provides the specified administration analytics.
19. The application works across desktop, tablet, and mobile layouts.
20. The complete core journey works from verification through resolution:

Verify → Protect → Report → AI Understands → Route → Track → Escalate → Prove → Resolve

25. FUTURE ENHANCEMENTS

The following features are outside the initial MVP or require further development:

- Official university student-information-system integration
- Official HEC integration
- Automatic external escalation through official integrations
- SMS services
- Real-time campus security tracking

- Large-scale deployment across multiple universities
- Dedicated mobile application
- Additional student-community features
- Advanced analytics

AI Opportunities

AI Opportunity	Where It Can Be Used	How AI Would Work	Benefit	Priority
Complaint Classification	Complaint submission	Analyze natural-language complaint and identify category/subcategory	Reduces manual categorization	Current/Applicable
Priority Detection	Complaint submission	Analyze issue severity and recommend priority	Helps identify urgent complaints	Current/Applicable
Smart Routing	Complaint management	Recommend responsible department/authority	Sends complaints to the appropriate team faster	Current/Applicable
Complaint Generation	Complaint submission	Convert informal student text into a structured complaint	Makes reporting easier and clearer	Current/Applicable
AI Complaint Assistant	Student complaint flow	Understand student questions and guide them toward appropriate complaint/support options	Simplifies the complaint process	Current/Applicable

AI Opportunity	Where It Can Be Used	How AI Would Work	Benefit	Priority
Student Card OCR	Verification	Extract university/student information from uploaded cards	Reduces manual verification effort	Current/Applicable
Policy/FAQ Assistant	Student support	Retrieve relevant approved policies, FAQs, student-rights information, and procedures before generating an answer	Provides easier access to relevant information	Current/Applicable
Complaint Pattern Analysis	Admin analytics	Analyze aggregated complaint data to identify recurring issues	Helps universities identify systemic problems	Future Enhancement
Duplicate/Similar Complaint Detection	Complaint management	Compare new complaints with existing complaints to identify similar issues	Can reduce duplicate handling	Future Enhancement

For the five-day MVP, the highest-value AI capabilities are complaint classification, priority recommendation, smart routing, structured complaint generation, AI assistance, and student-card OCR. Other AI opportunities should not delay completion of the core complaint-to-resolution workflow.